



UNIVERSITY<sup>AT</sup>ALBANY  
STATE UNIVERSITY OF NEW YORK

CSI 436/536 (Fall 2026)  
**Machine Learning**

Lecture 10: Regularization

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# Announcement

- HW 2: due tonight.
- Midterm exam: 2<sup>nd</sup> session next Tuesday Oct 6.
  - TA will review all solutions to HW 1 and 2 in the 1<sup>st</sup> session next Tuesday.
  - Come to our office hours!
- Midterm presentations on Tue Oct 20.
  - I'll give a quick tutorial on presentations today.

# Recap: Two problems of supervised learning

|                                       | Classification                                 |                                               | Regression        |
|---------------------------------------|------------------------------------------------|-----------------------------------------------|-------------------|
|                                       | Binary classification                          | Multi-class classification                    |                   |
| Feature space                         | $\mathbb{R}^d$                                 | $\mathbb{R}^d$                                | $\mathbb{R}^d$    |
| Label space                           | $\{-1, 1\}$                                    | $\{1, 2, 3, \dots, K\}$                       | $\mathbb{R}$      |
| Performance metric                    | Classification error (0-1 loss) for test data  | Classification error (0-1 loss) for test data | Mean Square Error |
| Popular surrogate loss (for training) | Logistic loss / exponential loss / square loss | Multiclass logistic loss (Cross-Entropy loss) | Square loss       |

# Recap: The objective function for learning linear regression under square loss

- $\hat{w} = \operatorname{argmin}_w \frac{1}{n} \sum_{i=1}^n (x_i^T w - y_i)^2 = \operatorname{argmin}_w \|Xw - y\|_2^2$ 
  - aka: Ordinary Least Square (OLS)
- Direct solver: setting gradient to 0.

# Recap: Derive the SGD algorithm

- Problem:
- $\hat{w} = \operatorname{argmin}_w \frac{1}{n} \sum_{i=1}^n (x_i^T w - y_i)^2 = \operatorname{argmin}_w \|Xw - y\|_2^2$
- Step 1: Calculate the gradient of the square loss
- Step 2: Write the SGD update rule

# Recap: How to solve linear regression?

- Challenges:
  - We don't have access to future data for prediction!
  - We also don't have access to ground truth
- By solving an optimization problem that **minimizes the loss function on the training data**, and hope that it generalizes.
  - We can verify if it generalizes or not using hold-out / cross-validation ...
- The least square optimization problem using square loss function:

$$\min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^n (f(x_i) - y_i)^2$$

# Recap: Linear regression

- Stochastic Gradient Descent (SGD)

- Using a **stochastic approximation** of the gradient:

$$\theta_{t+1} = \theta_t - \eta_t \hat{\nabla} f(\theta_t)$$

- Calculated by one data point randomly sampled from dataset

- Linear regression

- $\hat{w} = \operatorname{argmin}_w \frac{1}{n} \sum_{i=1}^n (x_i^T w - y_i)^2 = \operatorname{argmin}_w \|Xw - y\|_2^2$
  - Direct solver:  $\hat{w} = (X^T X)^{-1} X^T y$
  - GD:  $w \leftarrow w - 2\eta X^T (Xw - y)$
  - SGD:  $w \leftarrow w - 2\eta x_i (x_i^T w - y_i)$

# Recap: Time complexity

- Direct solver
  - $O(nd^2 + d^3)$
- GD:
  - $O(ndT)$
- SGD:
  - $O(dT)$
- $T = \text{n\_iterations}$

# Today

- Linear regression in curve fitting
- Recap: Problem of overfitting
- **Regularization prevents overfitting**
- Quick tutorial on course project presentation

# Example of regression - Curve fitting: How to train a function fitting blue dots?

- Input:

$$x \in \mathcal{X} = [0, 1] \subset \mathbb{R}$$

- Output:

$$y \in \mathcal{Y} = \mathbb{R}$$

- Data:

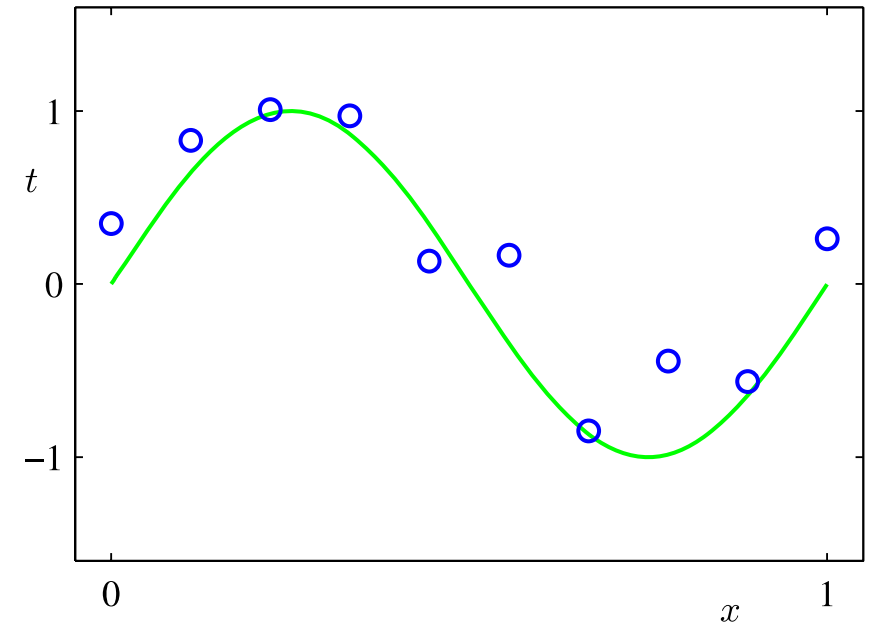
$$(x_1, y_1), \dots, (x_n, y_n)$$

- Ground truth:

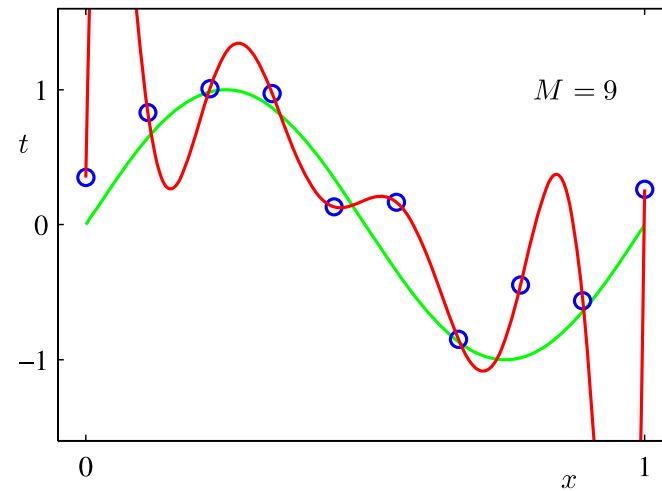
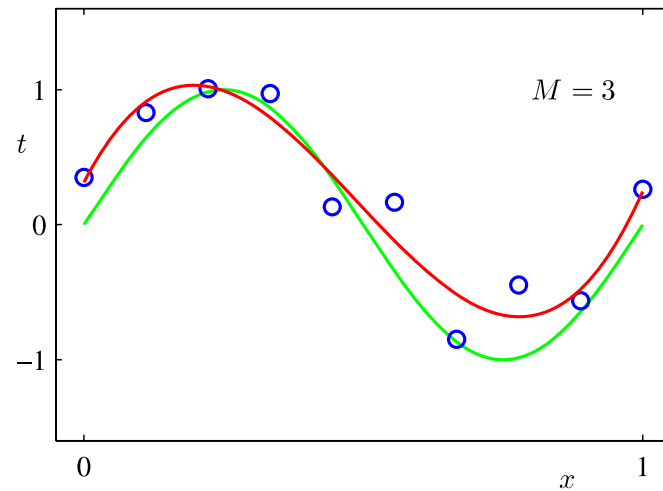
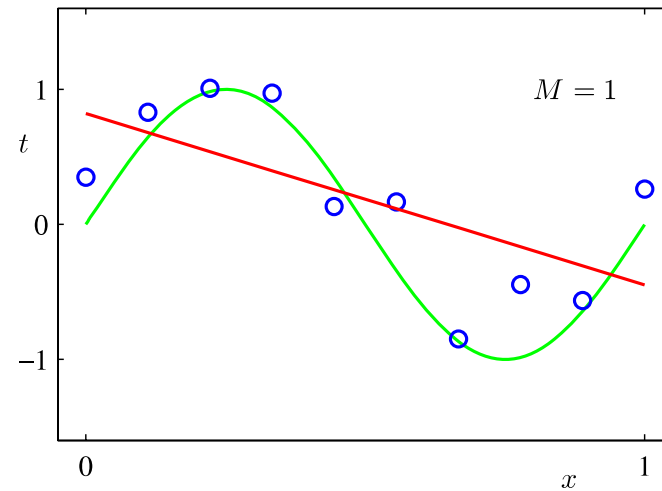
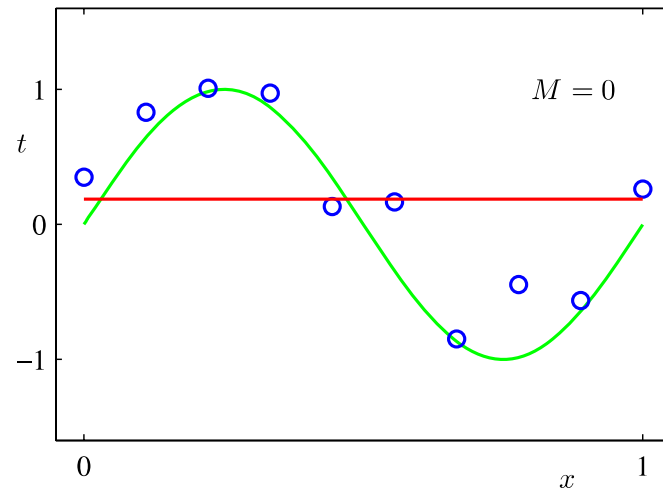
$$f_0(x) = \sin(2\pi x)$$

- Hypothesis (model)

$$f(x) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M$$

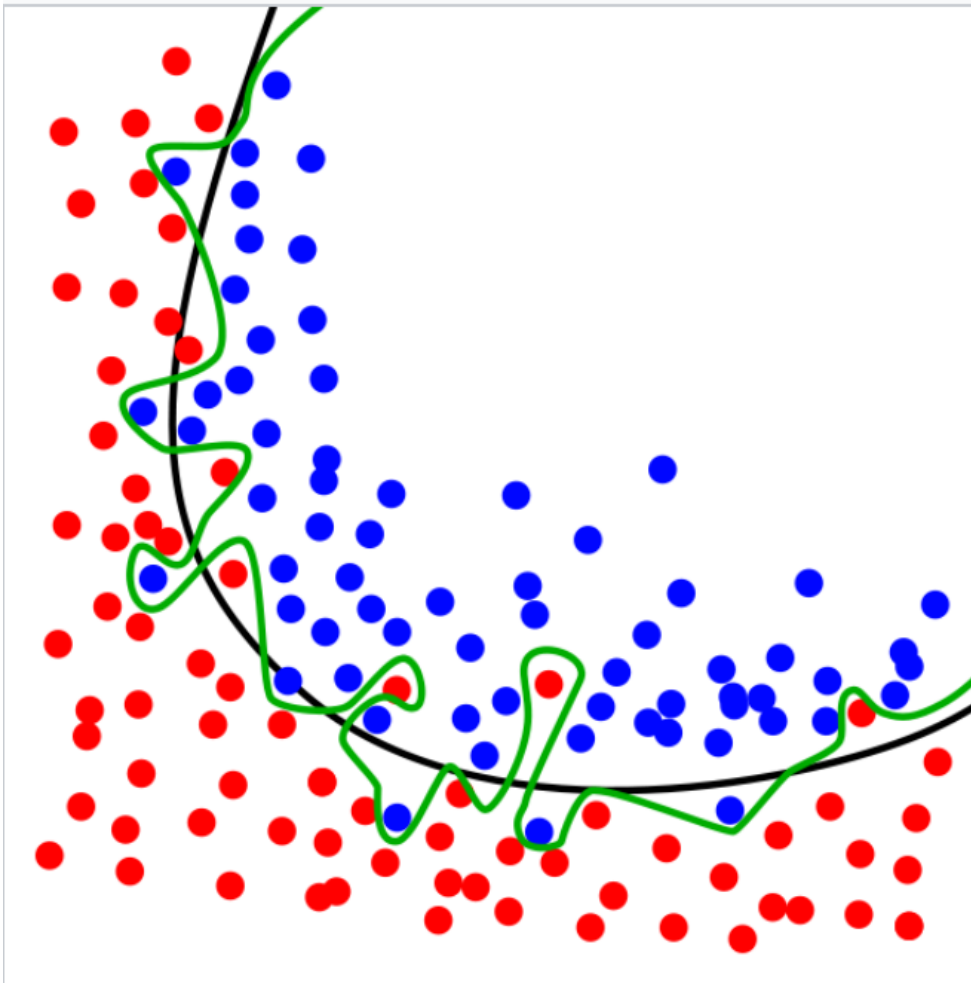


# Polynomial regression under square loss



Problem of overfitting!

# Recap: The problem of Overfitting



The **green** line represents an overfitted model.

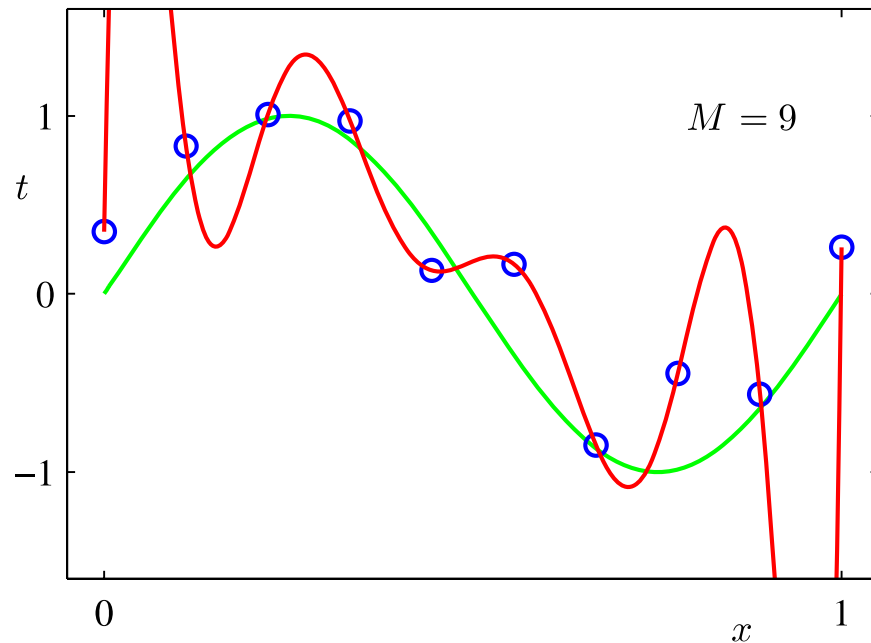
1. Best follows the training data
2. Too dependent on training data
3. More likely to fail (higher error rate) than black line on new unseen test data

Discussion: examples of overfitting in our learning as human beings?

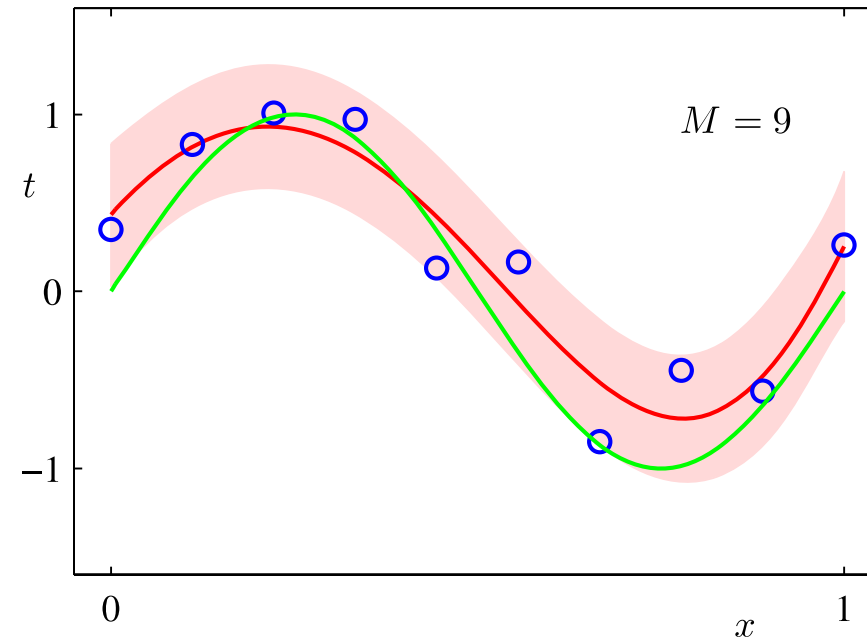
# Regularization prevents overfitting!

- Same model:

$$f(x) = w_0 + w_1x + w_2x^2 + \dots + w_Mx^M$$



No regularization



With regularization

# How does regularization work?

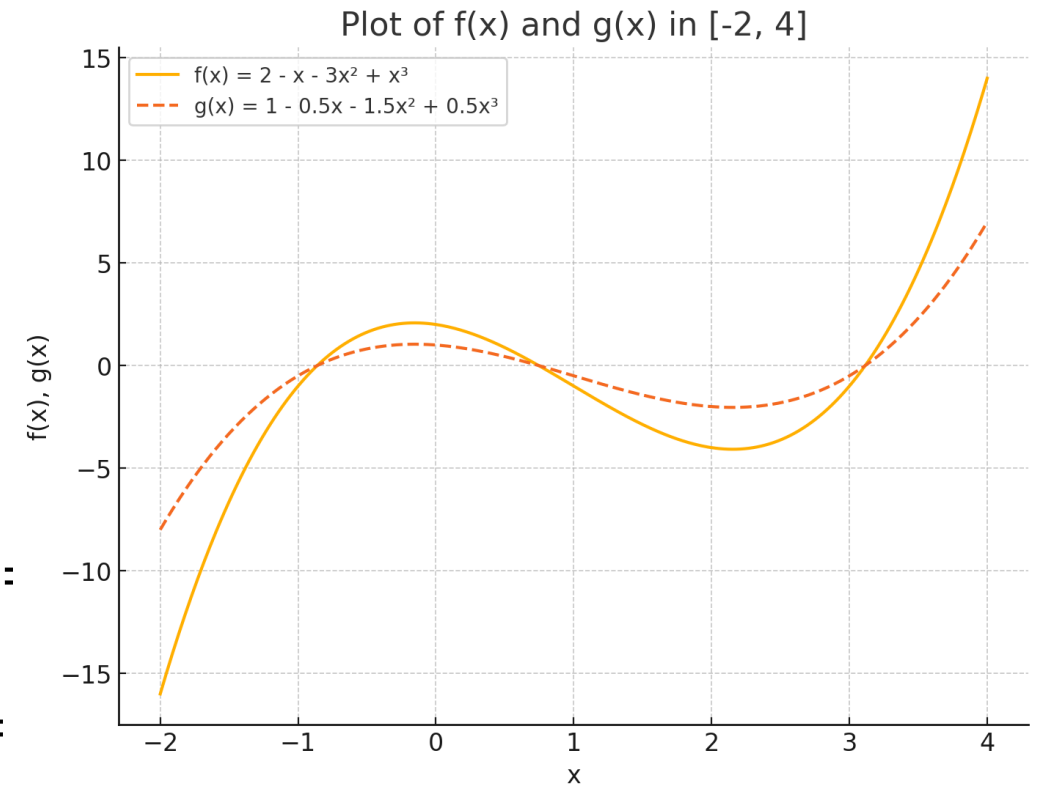
- Regularization controls the **parameter complexity** (norms of parameter).

- $p$ -norm regularized least square

$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \|X\theta - y\|_2^2 + \lambda \|\theta\|_p^p$$

- In-class exercise:

- Find L-2 norm of  $x = [2, -1, -3, 1]$  and  $x' = [1, -0.5, -1.5, 0.5]$
  - Plot  $f(x) = 2 - x - 3x^2 + x^3$  and  $g(x) = 1 - 0.5x - 1.5x^2 + 0.5x^3$  in  $[-2, 4]$



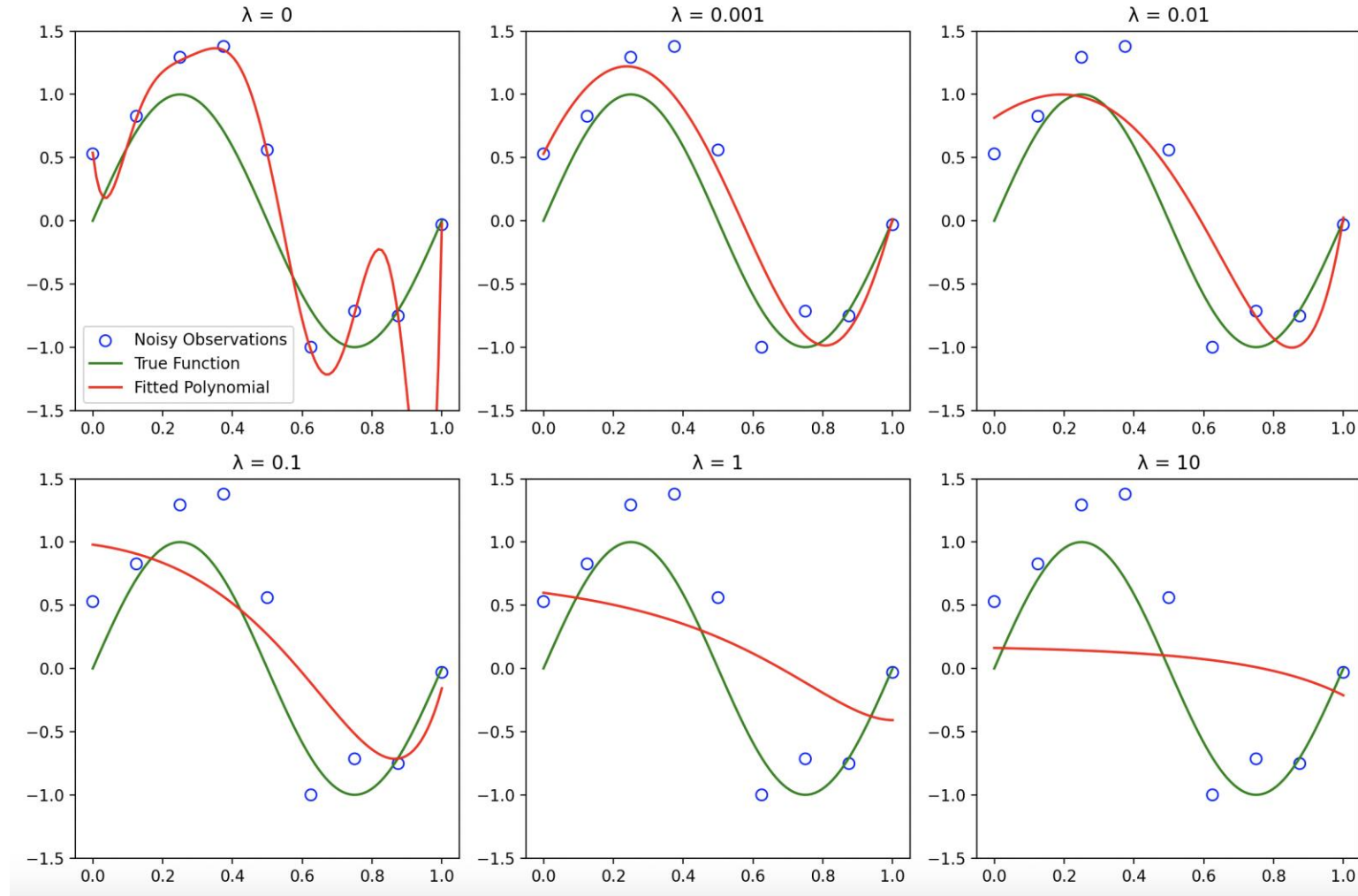
# Different choices of regularization

- p-norm regularized least square

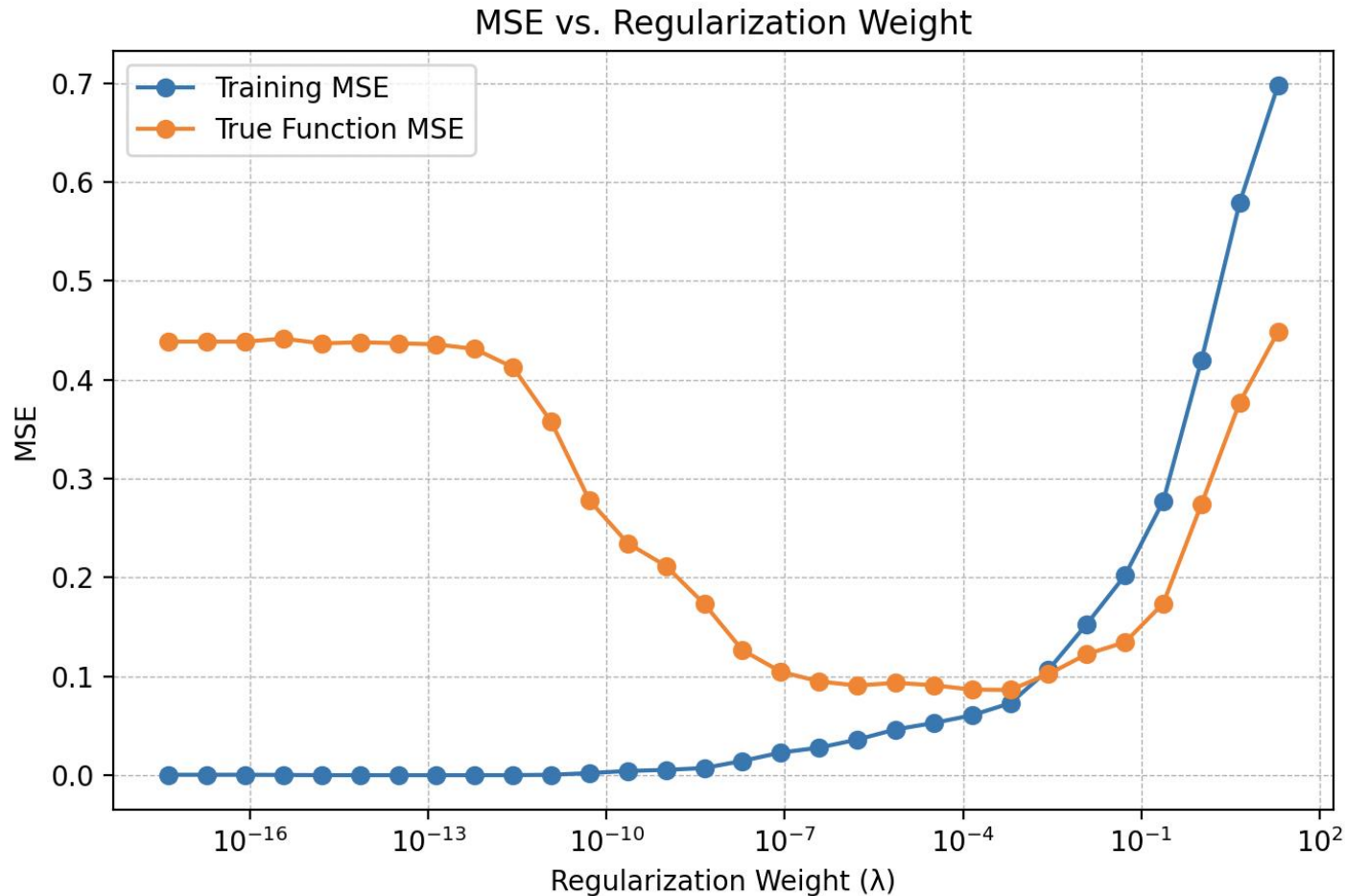
$$\hat{\theta} = \arg \min_{\theta} \frac{1}{n} \|X\theta - y\|_2^2 + \lambda \|\theta\|_p^p$$

- when  $p=2$ , this is called “**Ridge Regression**”
  - when  $p=1$ , this is called “Lasso”
- Direct solver for Ridge Regression in HW 2.

# Fitted curve as L-2 regularization weight increases



# The mean square errors as we adjust the L2 regularization weight



- Discussion:
  - Why **training error** increases with regularization weight?
  - Why is **test error** in a U-shaped curve?

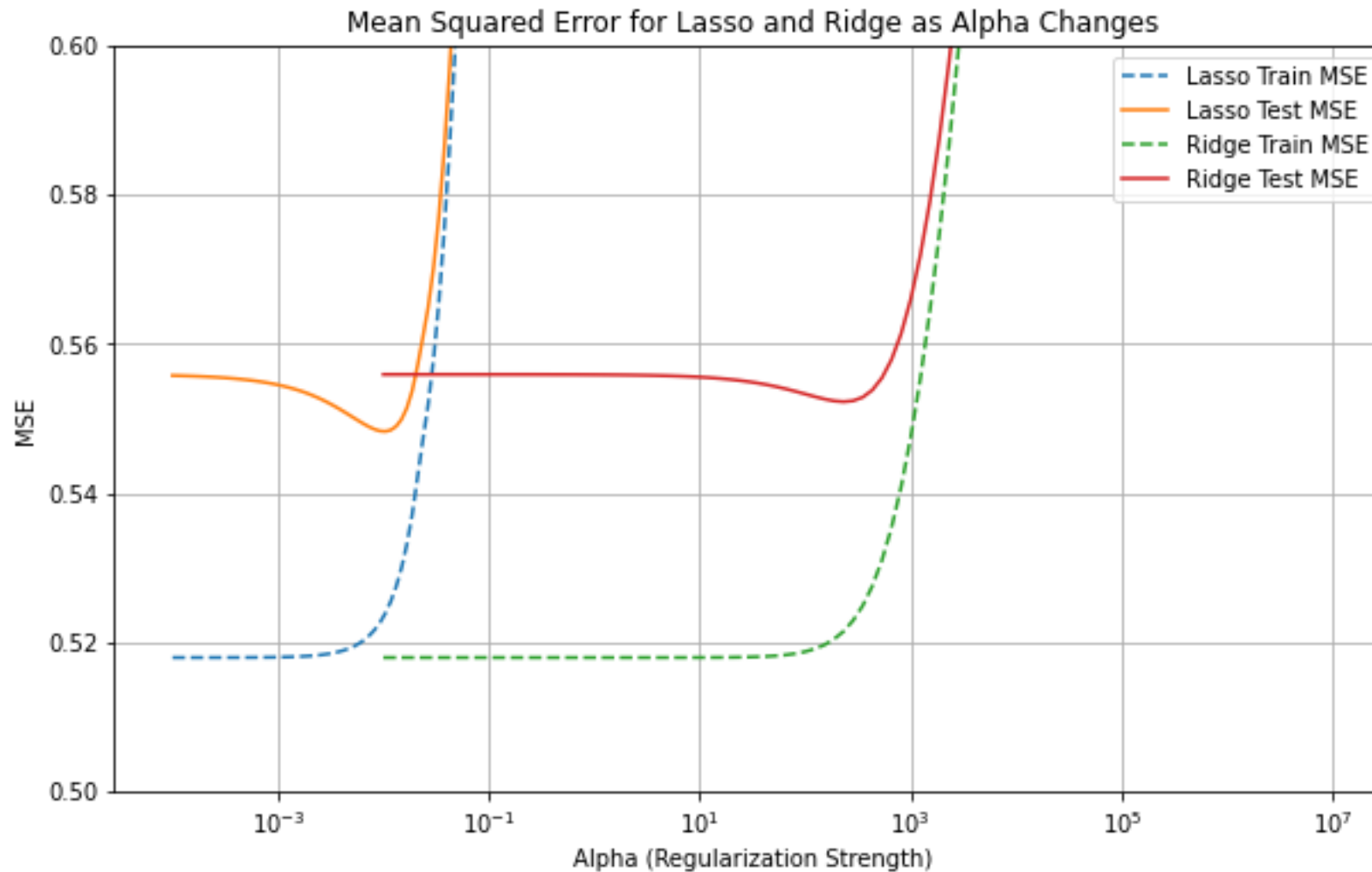
# Case study: Housing price dataset

- Example data

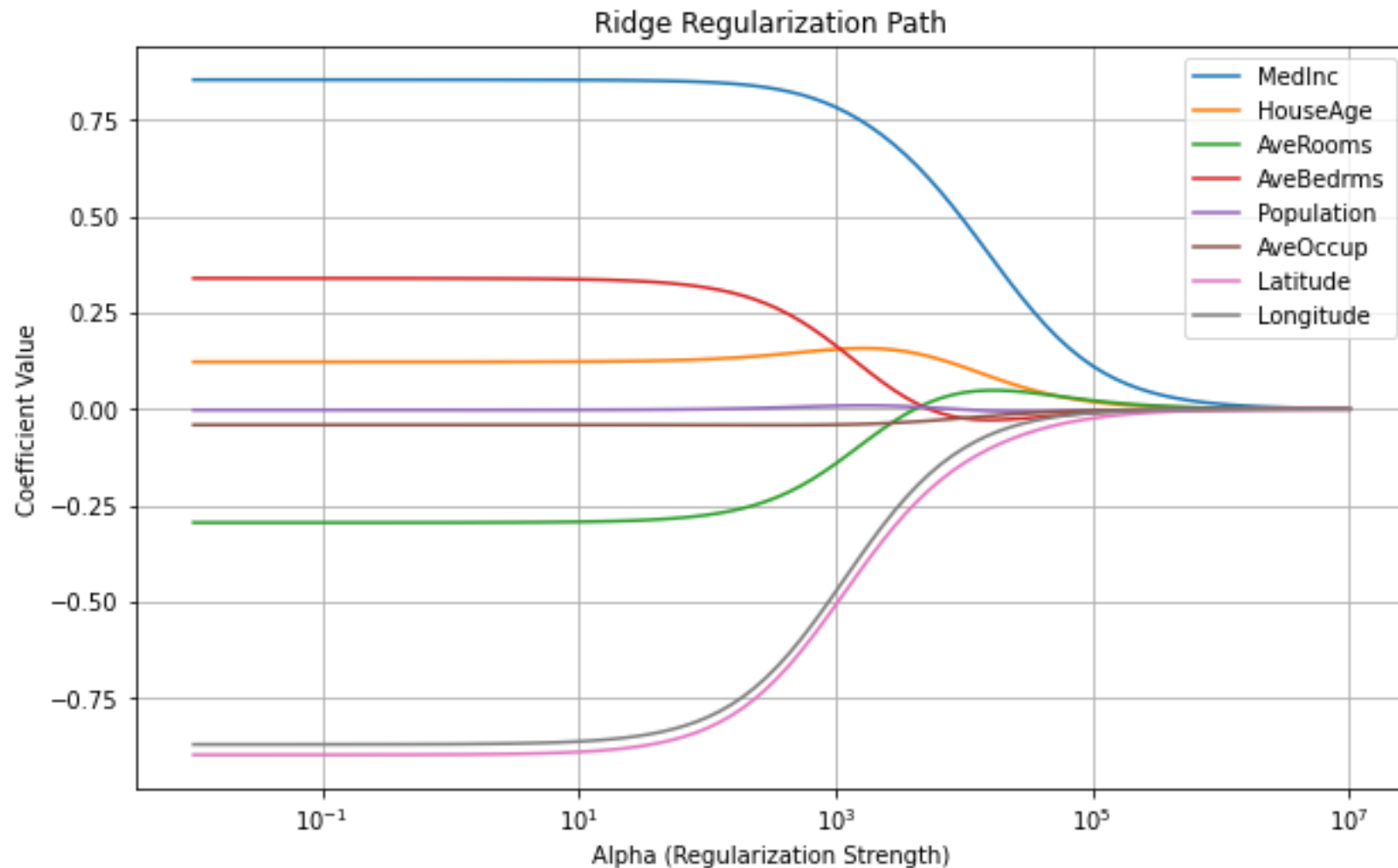
|   | MedInc | HouseAge | AveRooms | AveBedrms | Population | AveOccup | Latitude | Longitude | Target |
|---|--------|----------|----------|-----------|------------|----------|----------|-----------|--------|
| 0 | 8.3252 | 41.0     | 6.984127 | 1.023810  | 322.0      | 2.555556 | 37.88    | -122.23   | 4.526  |
| 1 | 8.3014 | 21.0     | 6.238137 | 0.971880  | 2401.0     | 2.109842 | 37.86    | -122.22   | 3.585  |
| 2 | 7.2574 | 52.0     | 8.288136 | 1.073446  | 496.0      | 2.802260 | 37.85    | -122.24   | 3.521  |
| 3 | 5.6431 | 52.0     | 5.817352 | 1.073059  | 558.0      | 2.547945 | 37.85    | -122.25   | 3.413  |
| 4 | 3.8462 | 52.0     | 6.281853 | 1.081081  | 565.0      | 2.181467 | 37.85    | -122.25   | 3.422  |

- Questions one can ask:
  - How well can one use the 8 features to predict house price (i.e., Target)?
  - Which feature is more predictive with house price?
  - What is the effect of regularization?

# The MSE vs regularization weight



# The “Regularization path” for L2-regularization



- The “sign” indicates positive or negative correlation with the label
- The “magnitude” indicates how strongly correlated.

# Summary

- Linear regression
  - Solving the Least Square problem {with GD, SGD and direct solver}
- Regularization
  - Controls the parameter complexity of the fitted function
  - Prevents overfitting!
  - Different regularization: L-2 (most popular) and L-1
- Case study: Predict Housing Price
  - Effect of regularization on training error and test error
  - Regularization path (Effect of regularization on coefficients)